Functional Connectotomy in Glioma: Reproduction of a Hopf Whole-Brain Model and Extension via Virtual Lesioning and Structural Perturbation

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We reproduced and extended the Hopf whole-brain model of Alexandersen, Douw, Zimmermann, Bick and Goriely for glioma-related changes in phase-lag index (PLI) functional connectivity. The model fits simulated PLI to empirical PLI by varying global coupling strength K and local excitability λ. We qualitatively reproduced the main findings of the original study: fits were organised by the normalised coupling $\kappa = K/\sqrt{\lambda}$, glioma required higher effective coupling than control, and tumour-region connectotomy reduced fitted coupling.

We then extended the analysis in two ways. First, we applied virtual lesioning to all 78 AAL regions to identify regions that influence the PLI fitting objective. Second, we directly weakened structural connections of selected virtual-lesioning drivers to test whether functional sensitivity corresponds to anatomical damage sensitivity. Virtual lesioning revealed spatially heterogeneous effects, partly associated with eigenvector centrality, but structural perturbation did not move control fits toward the glioma-like higher-coupling regime.

These results support the interpretation that glioma alters the modelled balance between local dynamics and interregional coupling, while showing that functional connectotomy should be interpreted as a sensitivity analysis of model fit rather than direct evidence of anatomical disconnection.

1. INTRODUCTION

Understanding how large-scale brain activity emerges from interactions between distributed neural populations is a central problem in computational neuroscience. Whole-brain models address this by combining local neural dynamics with large-scale structural connectivity, allowing functional connectivity patterns to be reproduced and interpreted mechanistically [5, 6]. A common approach is to represent brain regions as coupled oscillators, such that synchronisation between regions gives rise to coherent large-scale activity [3]. Among these models, Hopf oscillators are particularly useful because they capture the transition between stable and oscillatory dynamics in a low-dimensional normal form [7].

Alexandersen et al. [1] used a Hopf whole-brain model to investigate glioma-related alterations in brain dynamics. Fitting the model to phase-lag index (PLI) functional connectivity, they showed that the phase dynamics are governed primarily by a normalised coupling parameter reflecting the balance between interregional coupling and local excitability, rather than by either quantity independently. They further reported that this balance is shifted in glioma patients toward stronger effective global interactions, and that tumour-related regions contribute to this shift in connectotomy analyses. These results suggest that glioma alters not only local neural activity, but also the relative contribution of large-scale network interactions to whole-brain dynamics.

In this report, *fitted coupling* refers to the value of the model's global coupling parameter K that gives the best match between simulated and empirical PLI matrices. Because the model fit is governed by $\kappa = K/\sqrt{\lambda}$, fitted coupling should be interpreted as an effective model-level measure of interregional influence relative to local oscillatory amplitude, not as a direct measurement of anatomical white-matter strength. We also distinguish two forms of lesion analysis. *Functional connectotomy* or *virtual lesioning* removes a region from the PLI comparison only: the region remains present in the simulation, but its row and column are excluded when correlating simulated and empirical connectivity. *Structural perturbation* instead changes the simulated network itself by weakening the anatomical connections of a selected region before running the model.

In this project, we first reproduce the main dynamical findings of Alexandersen et al. [1]: the approximate $K/\sqrt{\lambda}$ scaling relation, the shift toward higher fitted coupling in glioma, and the reduction in fitted coupling after tumour-region connectotomy. We then extend the original connectotomy analysis in two targeted ways. The original paper focused on tumour-related region sets; our first extension applies the same masking logic systematically to every AAL region, producing a whole-brain atlas of region-wise coupling sensitivity. Our second extension asks whether these functional sensitivity results correspond to anatomical vulnerability by directly weakening the structural connections of selected functional driver regions before simulation. Together, these analyses distinguish regions that influence the fitting objective from regions whose structural disconnection is sufficient to reproduce glioma-like model behaviour. For clarity, Table I summarises only the hypotheses associated with the extension analyses.

2. METHODS

2.1. Hopf oscillator model

At each node, the dynamics were governed by the Hopf normal form

$$\dot{z}_i = z_i\left(\lambda + i\omega_i - |z_i|^2\right) + K\tanh\left(C\sum_{j=1}^{N} w_{ij}x_j\right), \quad (1)$$

where $z_i = x_i + iy_i$ is the complex state of node i, λ is the excitability parameter, ω_i is the intrinsic angular frequency, K is the global coupling strength, C is a structural scaling constant, and w_{ij} are the entries of the structural connectivity matrix. As in the original paper, the real part x_i was treated as the simulated neural signal. The implementation was first verified in single-node and two-node reductions before scaling to the empirical network (see Sec. 3.1).

2.2. Simulation pipeline

For each parameter setting, the coupled Hopf equations were numerically integrated from random initial conditions to generate one time series per brain region. After discarding an initial transient period, the real component $x_i(t)$ of each oscillator was retained as the simulated neural signal, following the original modelling framework. The simulated regional signals were then band-pass filtered in the alpha band and transformed to instantaneous phase using the Hilbert transform. For each pair of regions, phase-lag index (PLI) was computed as the absolute mean sign of the phase difference over time, giving one simulated PLI matrix for each simulation run. Model fit was quantified by taking the Pearson correlation between the upper-triangular entries of the simulated PLI matrix and the corresponding empirical PLI target. This pipeline was repeated across the parameter grid and, where stated, across multiple random initial conditions.

2.3. Network simulations

The final stage of the reproduction scaled the model to larger networks. Before fitting empirical data, synthetic network tests were used to confirm that the model produced plausible oscillatory dynamics and structured PLI matrices. In the empirical-scale simulations, the model was run on the 78-node AAL structural connectivity matrix using region-wise peak-frequency values from the empirical dataset. Control and glioma cohort-averaged PLI matrices were used as separate empirical fitting targets. Unless otherwise stated, simulations used the same processing pipeline described above: integration of the Hopf system, removal of an initial transient, extraction of $x_i(t)$, alpha-band filtering, Hilbert phase estimation, PLI computation, and simulated–empirical PLI correlation. For each pair of regions, PLI was computed as the absolute mean sign of the phase difference over time, so that values near zero indicate symmetric phase differences and values near one indicate a consistent non-zero phase lag [9].

2.4. Parameter fitting and comparison with empirical data

To reproduce the fitting procedure in the original paper, a two-dimensional grid search over coupling strength K and excitability λ was performed. For each parameter pair, the simulated PLI matrix was compared with the empirical PLI using the Pearson correlation between the upper-triangular entries of the matrices. Repeated simulations over multiple random initial conditions were used to obtain more stable estimates of fit quality. This fitting procedure was first used to reproduce the control-cohort parameter landscape corresponding to Figure 2 of the original paper. A shuffled-connectivity null model was then introduced by randomising the empirical structural connectivity matrix while preserving the weight distribution, in order to test whether the observed fit depended on the true network topology. Thresholded comparisons were also carried out using top-3% PLI matrices, matching the thresholding procedure implemented in the network simulations. Whereas Alexandersen et al. [1] masked predefined tumour-region sets, we apply the same masking systematically to each of the 78 AAL regions.

2.5. Scaling-law reproduction

To reproduce the scaling result reported in the original paper, we evaluated model fit over logarithmically spaced grids of global coupling strength K and excitability λ. For each parameter pair, the simulated PLI matrix was compared with the empirical PLI matrix using the same Pearson-correlation fitting criterion described above. The key quantity tested was the normalised coupling

$$\kappa = \frac{K}{\sqrt{\lambda}}. \quad (2)$$

This means that lines of constant normalised coupling satisfy

$$K = \kappa\sqrt{\lambda}. \quad (3)$$

Thus, along a constant-κ line, K is not fixed; instead, it increases in proportion to $\sqrt{\lambda}$. We examined whether regions of high model fit followed these constant-κ lines in the log–log parameter landscape. If so, this would indicate that the phase-based functional connectivity is governed primarily by the balance between coupling strength and local excitability, rather than by either parameter independently. This analysis formed the central reproduction of the scaling-law result.

2.6. Glioma fitting and connectotomy analyses

The later sections of the reproduction extended the same fitting framework to the glioma cohort. Separate

TABLE I. Hypotheses and expectations for the extension analyses only.

Analysis	**Hypothesis or expectation**	**Outcome**
Virtual lesioning atlas	Structurally important regions, especially hubs, should produce larger virtual-lesioning shifts in fitted coupling than peripheral regions.	Partially supported
Virtual lesioning atlas	Top glioma-specific virtual-lesioning drivers should overlap significantly with the tumour-region set used by Alexandersen et al. [1].	Partially supported
SC perturbation	Directly damaging top virtual-lesioning drivers should produce steeper K^* dose-response shifts than damaging non-driver regions.	Not supported
SC perturbation	Damaging top drivers in the control model should push fitted control coupling upward toward the glioma regime.	Not supported
Extension synthesis	Regions that are strong functional drivers in Virtual lesioning atlas should also be strong anatomical sensitivity drivers in SC perturbation (the "consistent hub" hypothesis).	Not supported

parameter searches were performed against the glioma PLI matrix, and the resulting optimal coupling strengths were compared with the control case across multiple values of λ. Distributions of optimal coupling differences were then computed over repeated initial conditions, together with Wilcoxon tests, to reproduce the cohort-level comparisons in the original study. Finally, functional connectotomy analyses were implemented to reproduce the paper's Figures 6–8. Tumour-related regions were removed from the PLI comparison stage, the model was refit, and the resulting changes in optimal coupling were analysed for tumour-region groupings and pooled distributions. This allowed the contribution of tumour regions to elevated coupling strength to be assessed within the reproduced modelling framework.

2.7. Virtual lesioning atlas

To extend the connectotomy analysis beyond the predefined tumour-region sets used in Alexandersen et al. [1], a systematic virtual-lesioning analysis was performed over all 78 AAL regions, based on the automated anatomical labelling parcellation [10]. The same spectral-normalised empirical structural connectivity matrix, raw control and glioma PLI matrices, and subject-mean cortical frequencies used in the reproduction were retained. Simulations were conducted at a fixed excitability value $\lambda = 40$, corresponding to the best-fit empirical value identified in the reproduction. For each virtual-lesioning run, the Hopf model was first simulated on the full, unchanged structural connectivity matrix. The lesion was applied only after simulation, at the PLI-comparison stage. Thus, the simulated dynamics still contained all 78 regions, but the selected region was excluded when computing the Pearson correlation between simulated and empirical PLI matrices. For each of 300 random initial conditions, the global coupling parameter K was swept over 60 linearly spaced values between 0.5 and 50. At each (K, λ) pair, the full 78-node Hopf model was simulated once using structural scaling constant $C = 20$, total simulated time $t_{\text{total}} = 8.0$, discarded transient time $t_{\text{discard}} = 1.5$, and sampling frequency $f_s = 300$. The real component $x_i(t)$ of each oscillator was retrained, filtered in the alpha band, transformed to phase using Hilbert transform, and converted into a simulated PLI matrix. This matrix was compared with the empirical control and glioma PLI matrices using the Pearson correlation between the upper-triangular matrix entries. To implement virtual lesioning, each region was removed in turn only from the comparison stage. Specifically, after the full-network simulated PLI matrix had been generated, the corresponding row and column were deleted from both the simulated and empirical PLI matrices, and the correlation was recomputed on the reduced matrices. No structural connections were changed and the region was not removed from the Hopf simulation itself. For each initial condition and each region i, the parameter sweep yielded an optimal full-network coupling K^*_{full} and a region-masked optimum $K^*_{\text{lesion},i}$, defined as the values of K that maximised the correlation with the empirical data. The lesioning effect of region i was then quantified as

$$\Delta K_i = K^*_{\text{lesion},i} - K^*_{\text{full}}, \tag{4}$$

computed separately for the control and glioma comparisons. Negative values of ΔK_i indicate that masking a region reduces the fitted optimal coupling, implying that the region contributes positively to the elevated coupling required to reproduce the empirical functional connectivity.

To isolate disease-specific effects, a differential lesioning score was defined as

$$\Delta K_i^{\text{diff}} = \Delta K_i^{\text{glioma}} - \Delta K_i^{\text{control}}. \tag{5}$$

This quantity captures the extent to which a region's contribution to coupling strength is specific to the glioma

cohort, rather than being a general feature of the model fit.

Per-region effects were summarised across initial conditions by their mean and the 2.5th and 97.5th percentiles. Statistical evidence for non-zero lesioning effects was assessed using Wilcoxon signed-rank tests across initial conditions, applied to the non-zero entries of each regional effect vector, with significance evaluated using a Bonferroni-corrected threshold over the 78 regional tests. Because these tests were performed across random initial conditions, they quantify robustness to simulation variability rather than subject-level population inference. The empirical PLI targets were cohort averages, so these statistics should not be interpreted as tests of between-subject variability.

To assess whether strong functional coupling drivers coincide with structurally important regions, three graph-theoretic measures were computed from the same spectral-normalised empirical structural connectivity matrix for each region: degree, weighted strength, and eigenvector centrality. Spearman rank correlations were used to compare these measures with the lesioning effect sizes, separately for control and glioma fits. Finally, enrichment of the top-ranked lesioning drivers within the tumour-region set used in Alexandersen et al. [1] was assessed using a hypergeometric overlap test for top-N sets with $N \in \{5, 10, 15, 20, 30\}$. For the overlap analysis, the tumour set was represented as 19 regions grouped into Frontal L, Temporal L, and Frontal R categories.

Together, these analyses produced a region-wise atlas of functional coupling drivers and tested whether the connectotomy effect can be explained by the structural importance of the affected regions.

2.8. Structural connectivity perturbation

As a second extension, we tested whether anatomical damage to regions identified by the virtual-lesioning atlas changes the fitted global coupling in the whole-brain Hopf model. In contrast to the virtual-lesioning analysis, which removed regions only from the PLI comparison stage, this analysis directly perturbed the structural connectivity matrix used in the simulations and then refit the model. This analysis therefore differs from virtual lesioning in the order of operations. In virtual lesioning, the full model is simulated first and a region is ignored only when scoring the PLI fit. In structural perturbation, the structural connectivity matrix is modified before the Hopf equations are integrated. The resulting simulated dynamics therefore arise from a damaged connectome.

For each selected region and damage level, a perturbed structural connectivity matrix was first constructed. The Hopf model was then simulated on this perturbed connectome using the same empirical inputs as before: spectral-normalised structural connectivity, raw control and glioma PLI targets, subject-mean cortical frequencies, fixed excitability $\lambda = 40$, and the 60-point coupling grid $K \in [0.5, 50]$. Regions for perturbation were selected from the control virtual-lesioning ranking. The 10 regions with the most negative mean control lesioning scores ΔK were defined operationally as *functional drivers*, because masking them in the virtual-lesioning analysis produced the largest downward shifts in fitted coupling. This term refers only to influence on the fitting objective and does not imply that these regions are causal biological drivers of glioma pathology. As a null comparison set, 5 *non-driver* regions were also selected as those with the smallest absolute control lesioning effects.

Idealised structural damage was implemented as a graded dose-response manipulation. For a chosen region and damage level

$$d \in \{0, 0.25, 0.5, 0.75, 1.0\},$$

all incoming and outgoing structural connections of that region were multiplied by $1-d$. Thus, $d = 0$ left the connectome unchanged, $d = 0.5$ halved the selected region's connections, and $d = 1$ fully disconnected the selected node from the structural network. The Hopf model was then simulated using this perturbed structural matrix, and the resulting simulated PLI was fitted to the empirical control and glioma PLI targets. For each region, damage level, and initial condition, the model was simulated across the full K grid while keeping λ fixed. We used 50 initial conditions for each region–damage combination. For each run, K^*_{control} and K^*_{glioma} were defined as the coupling values that maximised the Pearson correlation with the corresponding empirical PLI target. Dose-response sensitivity was summarised by fitting the slope of K^* against damage fraction, both over the full damage range and over the mild range $0 \leq d \leq 0.5$. This directly tested whether regions identified as important by post-simulation functional masking also behave as important anatomical damage sites when the network itself is altered before simulation.

3. RESULTS

3.1. Reduced-model validation

Before turning to the empirical whole-brain model, we verified that the numerical implementation reproduced the expected Hopf dynamics in reduced settings. For a single uncoupled oscillator, simulations recovered the canonical Hopf bifurcation: trajectories decayed to the origin for $\lambda < 0$, remained at the bifurcation boundary for $\lambda = 0$, and converged to stable limit-cycle oscillations for $\lambda > 0$. The steady-state amplitude followed the theoretical prediction $|z| = \sqrt{\lambda}$, confirming that the implementation correctly captures the dependence of local oscillatory amplitude on excitability. In a symmetric two-node system, increasing coupling strength promoted phase locking and increased the phase lag index (PLI), with synchronisation approximately organised by

the normalised coupling

$$\kappa = \frac{K}{\sqrt{\lambda}}. \tag{6}$$

Taken together, these reduced-model results served mainly as validation and provided intuition for the scaling structure examined in the empirical network model.

3.2. Empirical network fitting and scaling law

We next scaled the model to the empirical 78-node structural connectome and compared simulated phase-lag connectivity with the empirical control-cohort PLI matrix. The simulated PLI reproduced broad block-structured organisation also visible in the empirical data, indicating that the coupled Hopf network captures non-trivial large-scale structure rather than producing homogeneous synchrony (Fig. 1).

To quantify model performance, we performed a grid search over coupling strength K and excitability λ, using the Pearson correlation between simulated and empirical PLI as the goodness-of-fit measure. The best fits occurred for positive values of λ, and the high-fit region extended diagonally through parameter space (Fig. 2). In the log–log parameter search, the high-fit region followed an approximately diagonal band through (K, λ) space. This pattern is expected if model fit is organised by the normalised coupling $\kappa = K/\sqrt{\lambda}$, since lines of constant κ satisfy

$$K = \kappa\sqrt{\lambda}.$$

This supports the interpretation that phase-based functional connectivity is governed primarily by the balance between interregional coupling and local excitability, rather than by K and λ independently.

3.3. Glioma-control comparison

We then compared the fitted parameter landscapes for the control and glioma cohort averages. Across excitability values, the glioma cohort was shifted toward higher optimal coupling strengths than the control cohort (Fig. 3). This difference was also reflected in the fitted normalised-coupling lines, which gave a larger effective κ for glioma than for control. The fitted normalised coupling was $\kappa_{\text{control}} = 5.460$ for the control cohort and $\kappa_{\text{glioma}} = 6.046$ for the glioma cohort, corresponding to an approximately 10.7% increase in the glioma fit. Thus, the reproduced whole-brain model supports the original paper's central claim that glioma-related dynamics are associated with stronger interregional interactions relative to local oscillatory dynamics.

3.4. Functional connectotomy

Finally, we reproduced the functional connectotomy analysis by removing tumour-related regions from the PLI comparison stage and refitting the model. In the glioma fit, masking tumour-region sets generally reduced the absolute fitted coupling, consistent with the direction of the original connectotomy effect. However, the control-relative change was heterogeneous across tumour sets: some tumour masks produced larger reductions in the control comparison than in the glioma comparison. We therefore interpret this as a qualitative reproduction of the connectotomy direction rather than an exact patient-level replication of the original effect.

3.5. Virtual lesioning atlas

To extend the connectotomy analysis beyond the predefined tumour-region sets used in the original paper, we evaluated the lesioning effect of all 78 AAL regions on the fitted optimal coupling. The resulting region-wise atlas showed that lesioning effects were spatially heterogeneous rather than uniform across the brain (Fig. 5). In the glioma fit, a subset of regions produced noticeably larger changes in the optimal coupling than the rest, whereas the control profile was comparatively flatter when shown in the same regional order. This suggests that the elevated coupling required to reproduce the glioma functional connectivity is driven disproportionately by specific regions rather than being distributed evenly across the network. To isolate disease-specific effects, we compared the lesioning responses in glioma and control using the differential score

$$\Delta K_i^{\text{diff}} = \Delta K_i^{\text{glioma}} - \Delta K_i^{\text{control}}.$$

The resulting differential atlas identified a subset of regions whose masking reduced the fitted coupling more strongly in glioma than in control (Fig. 6). The strongest positive differential effects were concentrated mainly in frontal and medial regions, including bilateral medial orbital and superior medial frontal areas, together with posterior cingulate and a smaller number of occipital and parietal regions. In contrast, several sensorimotor, insular, and subcortical regions showed negative differential effects, indicating weaker or more control-like lesioning responses. Taken together, these results move beyond the original grouped tumour-region connectotomy analysis by producing a systematic all-region sensitivity map of candidate functional coupling drivers.

3.6. Relation to structural connectivity

We next asked whether the strongest virtual-lesioning effects could be explained by the structural importance of the affected regions. Degree and weighted strength were not significantly associated with lesioning effect size in either cohort (control: degree $\rho = 0.07$, $p = 5.28 \times 10^{-1}$; strength $\rho = -0.04$, $p = 7.36 \times 10^{-1}$; glioma: degree $\rho = 0.01$, $p = 9.55 \times 10^{-1}$; strength $\rho = -0.05$, $p = 6.81 \times 10^{-1}$; Fig. 7). By contrast, eigenvector centrality showed a significant positive association with lesioning effect size in both cohorts, with a stronger relationship in glioma than in control (control: $\rho = 0.32$,

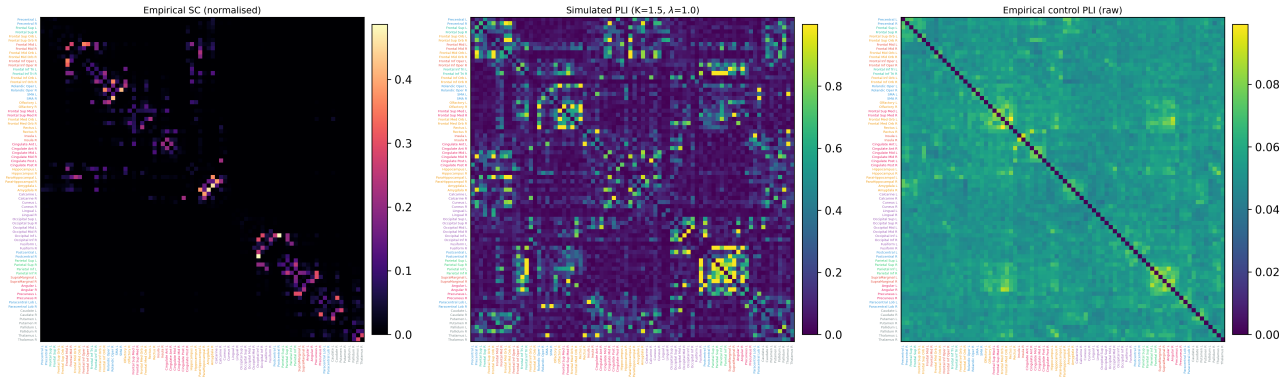

FIG. 1. Empirical-scale network fitting. Shown are the empirical structural connectivity matrix, a best-fit simulated PLI matrix, and the empirical control PLI matrix. The simulated matrix captures broad block-structured organisation also visible in the empirical data, supporting the use of the coupled Hopf network as a coarse-grained model of large-scale functional connectivity.

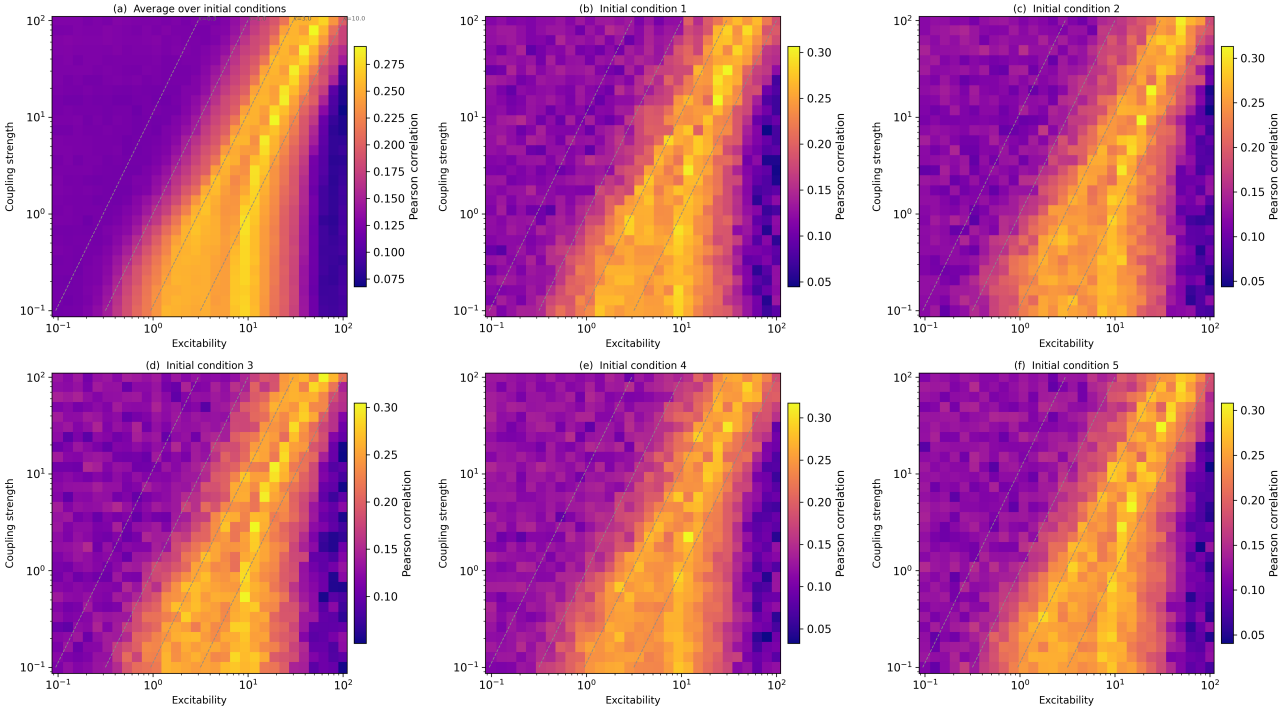

FIG. 2. Log–log parameter search for the empirical network model. Across multiple initial conditions, the best-fit region follows approximately diagonal bands corresponding to constant $\kappa = K/\sqrt{\lambda}$, indicating that the goodness of fit is largely controlled by the normalised coupling rather than by K and λ independently.

$p = 4.51 \times 10^{-3}$; glioma: $\rho = 0.41$, $p = 1.94 \times 10^{-4}$; Fig. 7). Thus, in this analysis, the strongest functional coupling drivers are not simply the nodes with the largest number or weight of connections, but are more closely related to globally influential regions in the structural network. Overall, the extension suggests that the connectotomy effect reported in the original paper is not only recoverable at the level of grouped tumour regions, but can be resolved into a whole-brain atlas of region-specific coupling drivers, with the clearest glioma-specific effects emerging in a subset of structurally influential frontal and medial regions. The overlap analysis showed that the top-ranked differential virtual-lesioning drivers were enriched for the tumour-region set used in the original paper at small N: the top-5 differential drivers contained 4 tumour-set regions ($p = 0.011$, hypergeometric test). However, this enrichment weakened for larger top-N sets, indicating that the overlap is concentrated among the strongest glioma-specific drivers rather than being a uniform property of the full ranking. As a parameter-

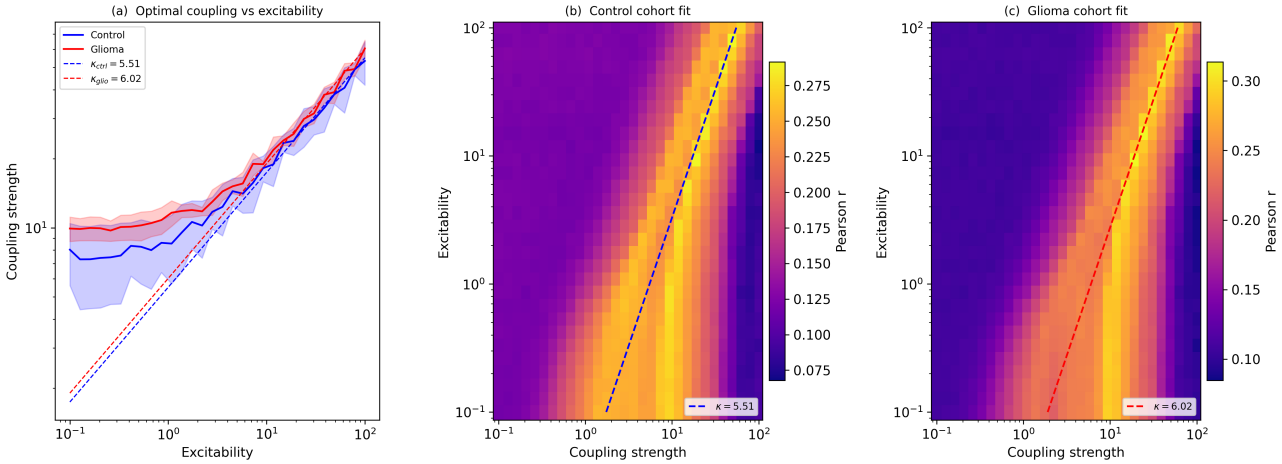

FIG. 3. Comparison of fitted parameter landscapes for the control and glioma cohorts. The optimal coupling curve lies systematically above the control curve, corresponding to a larger fitted normalised coupling for the glioma cohort. This indicates a shift toward stronger global interactions in the glioma network dynamics.

free comparator, ranking all 78 regions by the eigenvector centrality of the spectral-normalised SC alone—with no Hopf simulation, PLI fit, or grid search—recovered 7 of 19 tumour-set regions in the top-10 ($p = 1.4\times10^{-3}$) and 9 of 19 in the top-15 ($p = 1.1\times10^{-3}$), remaining significant at top-20 ($p = 1.7\times10^{-2}$, hypergeometric). The model-based differential ranking is therefore matched, not surpassed, by this topology-only comparator at every tested N.

3.7. Structural connectivity perturbation

We next tested whether directly damaging the structural connections of the control coupling-driver regions identified by the virtual-lesioning atlas would alter the fitted coupling regime. The top 10 control drivers included left posterior cingulate cortex, bilateral medial orbitofrontal cortex, left superior medial frontal cortex, and left inferior orbitofrontal cortex, and were contrasted with 5 non-driver regions whose lesioning effects were close to zero. At baseline ($d = 0$), the fitted coupling was consistently higher for the glioma target than for the control target across all tested regions, reproducing the cohort shift seen in the main analysis. For the driver set, mean fitted values were typically $K^*_{\text{control}} \approx 34$–$35$ and $K^*_{\text{glioma}} \approx 36$–$37$, with similar baseline values in the non-driver set. The full dose-response curves are shown in Fig. 10. Progressive structural damage did change the fitted coupling, but not in the hypothesised driver-specific direction. Across the mild damage range ($0 \le d \le 0.5$), the slope of K^*_{control} versus damage had mean -0.752 for driver regions and mean -1.946 for non-driver regions. Over the full range ($0 \le d \le 1$), the corresponding mean slopes were -2.583 for drivers and -2.497 for non-drivers. Thus, damaging the top coupling drivers did not produce larger or steeper coupling shifts than damaging the null regions; if anything, the mild-range effect was numerically stronger in the non-driver set. A one-sided Mann–Whitney U test on absolute mild-range slopes (drivers > non-drivers) was not significant ($U = 26$, $p = 0.48$), so the perturbation analysis did not support the prediction that virtual-lesioning driver status would translate into heightened structural sensitivity (Fig. 11).

To compare the two extension analyses directly, we next tested whether regions that were important in the virtual-lesioning atlas were also sensitive to direct structural perturbation. Functional importance was quantified as the absolute control virtual-lesioning effect, $|\Delta K_{\text{control}}|$, while structural sensitivity was quantified as the slope of K^*_{control} over the mild damage range $0 \le d \le 0.5$. Across the 15 perturbed regions, virtual-lesioning importance was not significantly associated with structural-perturbation sensitivity (Spearman $\rho = 0.15$, $p = 0.594$; Fig. 8). The same conclusion held when structural sensitivity was computed over the full damage range ($\rho = -0.22$, $p = 0.420$). Thus, regions whose masking strongly affected the PLI fitting objective were not necessarily the regions whose anatomical weakening most strongly altered the simulated network dynamics.

Consistent with this separation between functional and structural sensitivity, the perturbation also failed to push the control fit toward the glioma-like coupling regime. Averaged across driver regions, the control-glioma gap in fitted coupling was $\langle K^*_{\text{control}}\rangle - \langle K^*_{\text{glioma}}\rangle = -1.629$ at baseline, -1.770 at $d = 0.5$, and -3.584 at full damage. The corresponding non-driver gap changed from -1.339 at baseline to -1.832 at $d = 0.5$ and -2.883 at full damage. Rather than converging toward the glioma fit, the control fit therefore moved further away from it as damage increased, especially under extreme ablation (Fig. 9).

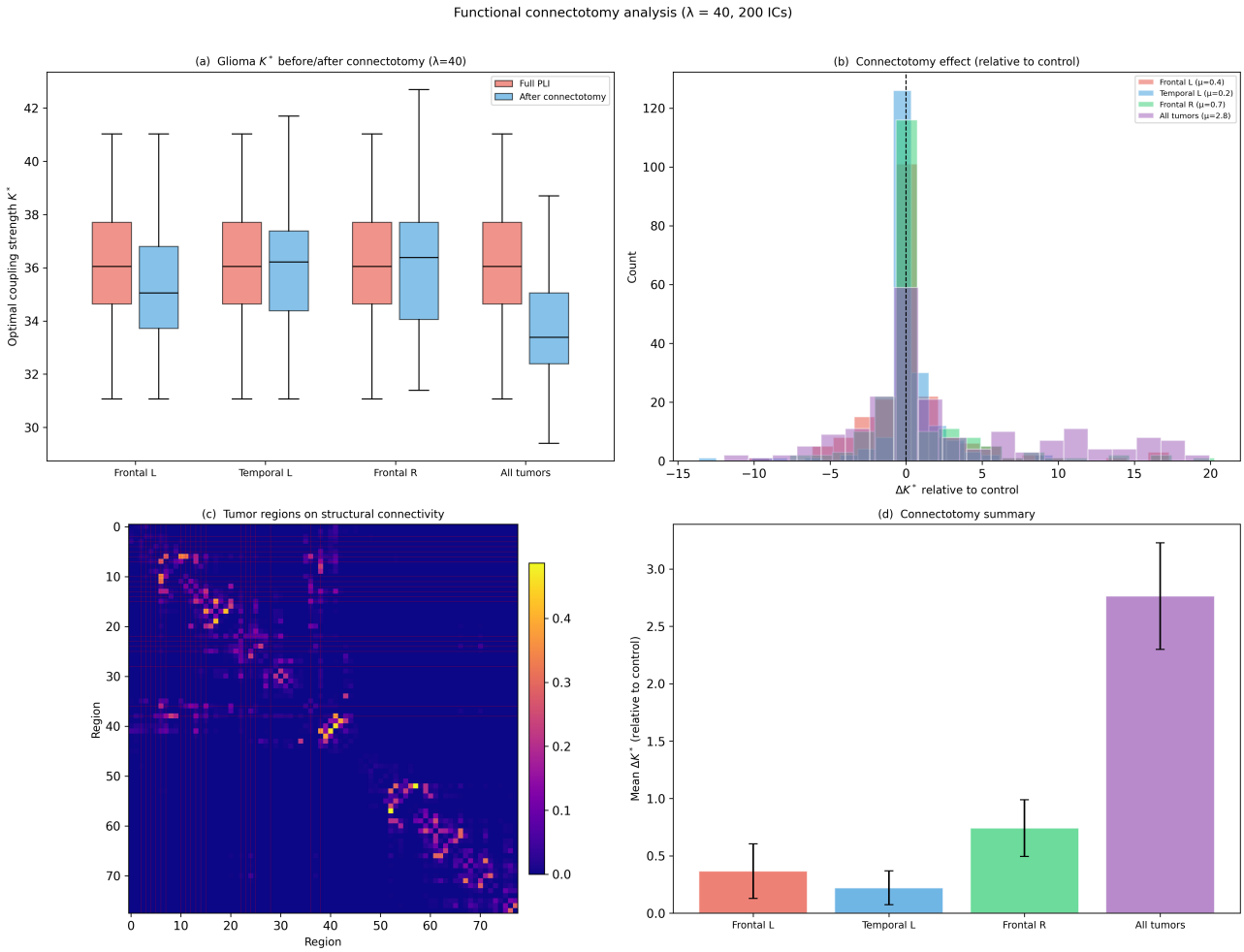

FIG. 4. Functional connectotomy analysis. Tumour-related regions were removed from the simulated–empirical PLI comparison and the model was refitted. In the glioma target, masking tumour-region sets generally reduced absolute fitted coupling, but the control-relative effect was heterogeneous across tumour groupings. This supports a qualitative reproduction of the original connectotomy direction while showing that the cohort-averaged connectotomy effect is not uniform across region sets.

Taken together, these results indicate that direct structural damage alone was not sufficient to reproduce the elevated-coupling signature associated with glioma. The perturbation manipulation altered fitted coupling, but it did not selectively amplify the effect of the strongest functional drivers and did not generate control-to-glioma convergence. Within this implementation, the glioma shift therefore cannot be explained as a simple consequence of weakening the structural connectivity of the virtual-lesioning analysis driver regions.

4. DISCUSSION

4.1. Reproduction of the original model

This project qualitatively reproduced the central dynamical findings of Alexandersen et al. [1] and extended their connectotomy framework using region-wise virtual lesioning and structural-connectivity perturbation. At the reproduction level, the Hopf model recovered the main structure of the original result: fits were organised primarily by the normalised coupling $\kappa = K/\sqrt{\lambda}$, glioma required higher effective coupling than control, and tumour-region connectotomy reduced fitted coupling in the expected direction. Because the fit is organised by κ, the glioma–control gap is best read as a shift in effective dynamical balance, requiring stronger interregional influence relative to local oscillatory amplitude rather than as a direct change in anatomical connectivity.

4.2. Region-wise virtual lesioning

The virtual-lesioning extension showed that fitted coupling sensitivity is spatially heterogeneous. By masking each of the 78 AAL regions from the PLI comparison, we obtained a whole-brain atlas of regions that influence the fitting objective. The strongest glioma-specific effects were concentrated mainly in frontal and medial regions, and the small-N tumour-overlap result suggests

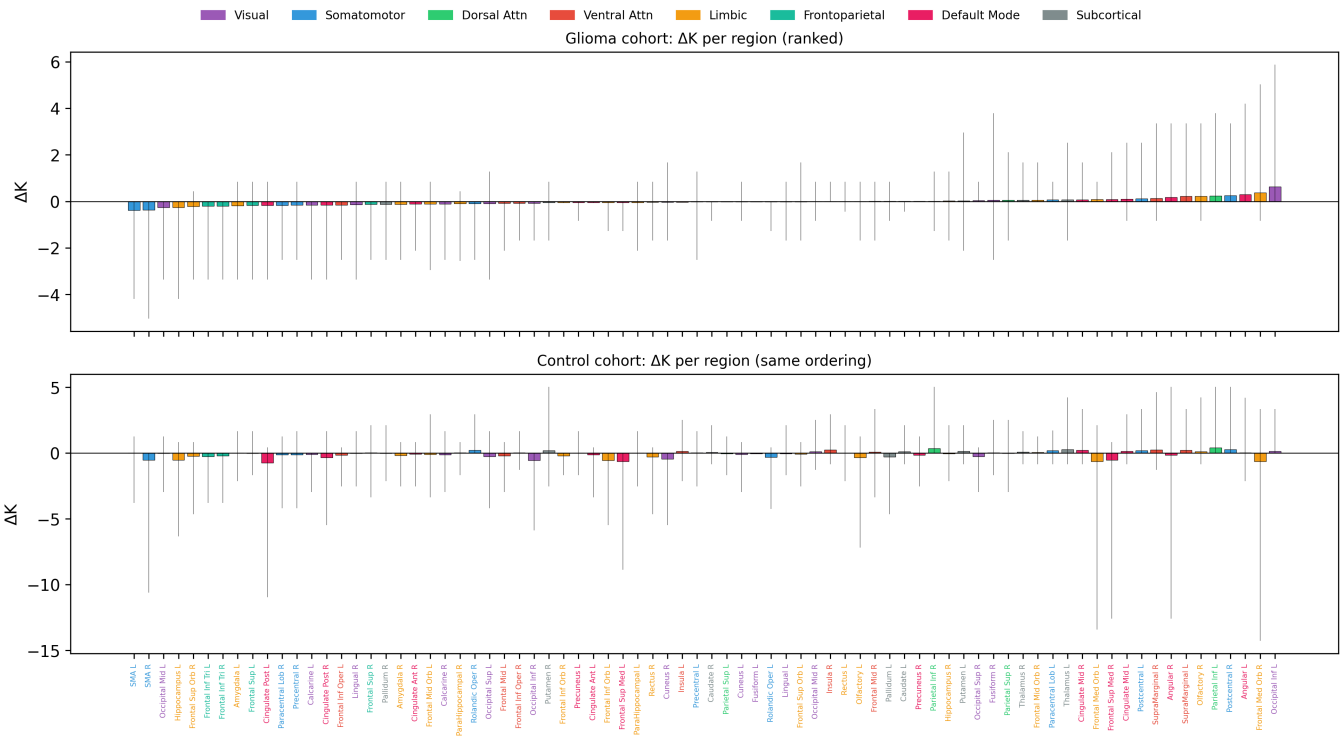

FIG. 5. Region-wise virtual-lesioning effects on fitted coupling. Top: glioma cohort lesioning effect ΔK for all 78 AAL regions, ranked by effect size. Bottom: control-cohort lesioning effects shown in the same regional order. Regions are coloured by large-scale network affiliation. The atlas shows spatially heterogeneous fitted-coupling sensitivity, with several glioma regions showing larger mean effects than the rest of the ranking.

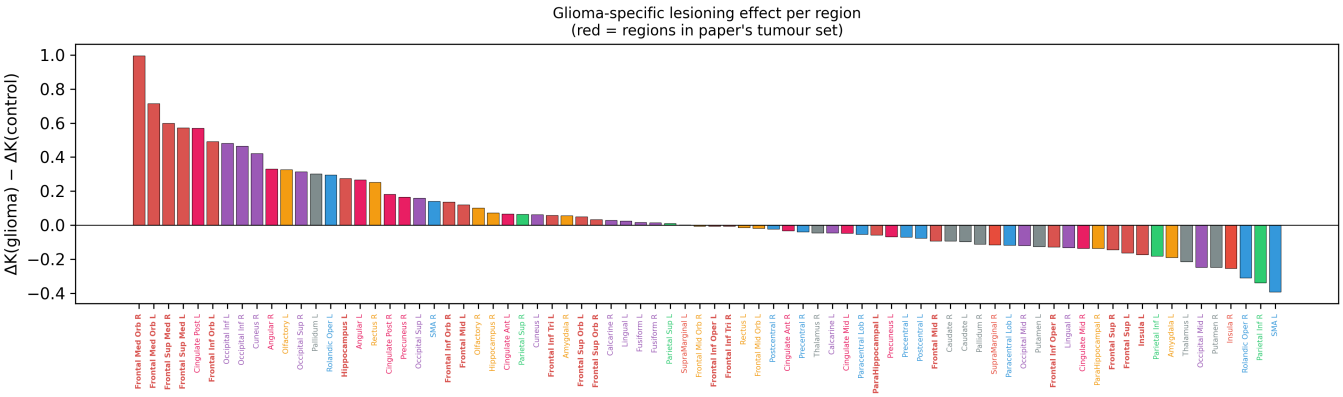

FIG. 6. Glioma-specific virtual-lesioning effect per region, quantified by $\Delta K_i^{\text{glioma}} - \Delta K_i^{\text{control}}$. Red labels indicate regions belonging to the tumour-region set used in the original paper. Positive values indicate regions whose masking reduces fitted coupling more strongly in glioma than in control.

that the strongest differential drivers partly coincide with the tumour-region set used in the original paper. The association with eigenvector centrality, but not degree or weighted strength, further suggests that these effects depend more on global structural network position than on simple connection count or total connection weight, consistent with graph-theoretic approaches to structural and functional brain networks [2]. An important caveat is that a parameter-free baseline ranking regions by eigenvector centrality of the spectral-normalised SC alone matches or exceeds the model-based differential ranking on the tumour-overlap test across every tested top-N. This suggests that, in the present implementation, the Hopf model adds dynamical content to the cohort offset—the elevated κ in glioma—rather than spatial information about which regions the differential effect localises

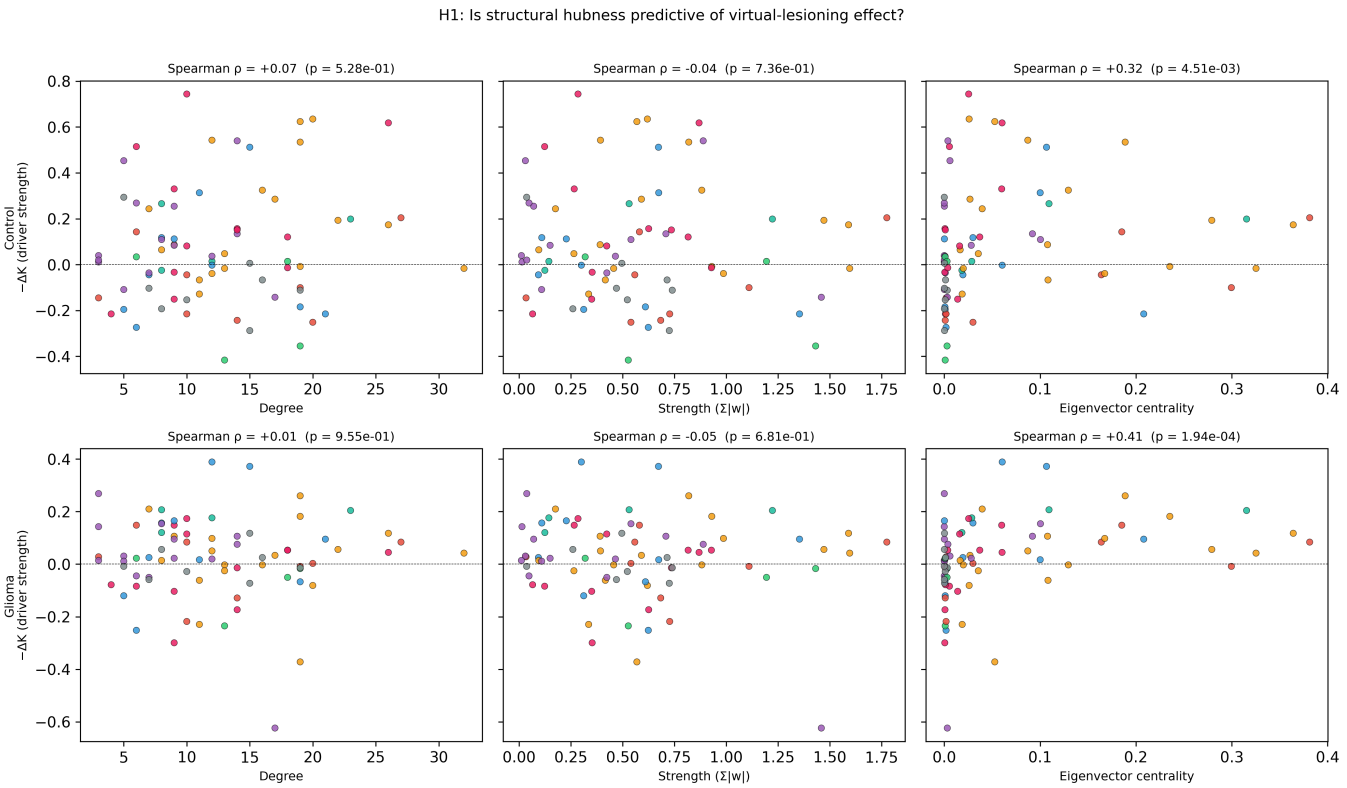

FIG. 7. Relationship between structural hubness and virtual-lesioning effect size. Degree and weighted strength show no significant association with lesioning effects, whereas eigenvector centrality is positively associated with lesioning effect size in both control and glioma fits, with a stronger relationship in glioma.

to, which is largely fixed by SC topology once spectral normalisation is applied.

4.3. Structural perturbation

The structural-connectivity perturbation analysis provided an important qualification. Unlike virtual lesioning, which removes a region only from the PLI comparison, structural perturbation directly weakens that region's anatomical connections before simulation. If virtual-lesioning drivers were also anatomical damage drivers, weakening them should have moved control fits toward the higher-coupling glioma regime. This was not observed: driver regions did not show stronger damage sensitivity than non-drivers, and structural damage generally increased rather than reduced the control–glioma coupling gap. The direct comparison between the two extension analyses further supported this interpretation: regions with larger virtual-lesioning effects did not show larger structural-perturbation slopes, indicating that functional fit-sensitivity and anatomical damage sensitivity are separable in this model. The dissociation is also visible at the level of individual regions: left posterior cingulate, the strongest functional driver from the virtual-lesioning atlas, was among the *least* sensitive regions under structural perturbation, whereas left precuneus—weak under virtual lesioning—was among the most sensitive. More broadly, this implies that the cost-function sensitivity of a fitting objective should not be read as an anatomical-importance score: the two quantities are formally distinct and, in this dataset, are empirically uncorrelated.

4.4. Overall interpretation

Together, the extensions show that functional connectotomy is best understood as a sensitivity analysis of the fitting objective, not as a direct simulation of tumour-induced anatomical disconnection. The virtual-lesioning atlas identifies regions that influence the empirical–simulated PLI comparison, whereas the perturbation analysis tests whether anatomical weakening changes the simulated network in a glioma-like direction, motivated by evidence that resting-state functional connectivity is constrained by underlying structural connectivity [4, 8]. The results therefore support the original interpretation that glioma alters the modelled balance between local oscillatory dynamics and interregional coupling, while showing that functional lesioning drivers do not necessarily behave as structural damage drivers.

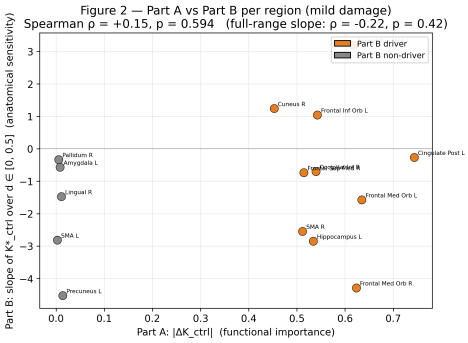

FIG. 8. Relationship between virtual-lesioning importance and structural-perturbation sensitivity. Each point represents one of the 15 regions tested in the structural perturbation analysis. The x-axis shows functional importance from the virtual-lesioning atlas, quantified as $|\Delta K_{\text{control}}|$. The y-axis shows structural sensitivity, quantified as the slope of fitted K^*_{control} over the mild damage range $0 \leq d \leq 0.5$. Virtual-lesioning importance did not significantly predict structural-perturbation sensitivity, indicating that functional fit-sensitivity and anatomical damage sensitivity capture distinct properties of the model.

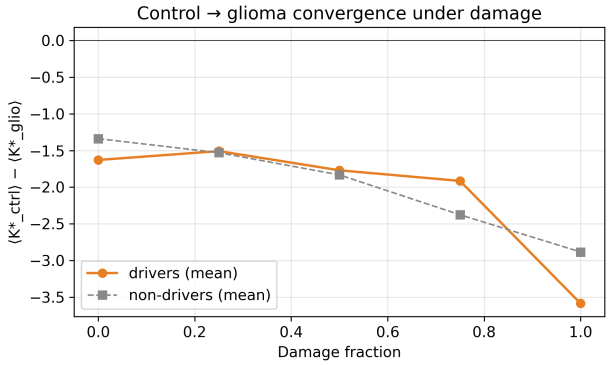

FIG. 9. Change in the control-glioma coupling gap under structural damage. Plotted is the mean difference $\langle K^*_{\text{control}} \rangle - \langle K^*_{\text{glioma}} \rangle$ across driver and non-driver groups as a function of damage fraction. More negative values indicate that the control fit lies further below the glioma fit. Damage increases this separation rather than producing convergence.

4.5. Limitations

Several limitations should be considered when interpreting these results. First, statistical tests performed across random initial conditions quantify simulation robustness rather than subject-level population inference, since the empirical targets were cohort-averaged PLI matrices and a single control structural connectome was used for both cohorts, so tumour-induced structural changes are not modelled. Second, the Hopf whole-brain model is a simplified phenomenological description of neuronal dynamics, so fitted parameters should be interpreted as model-level descriptors rather than direct physiological quantities, and conclusions are conditional

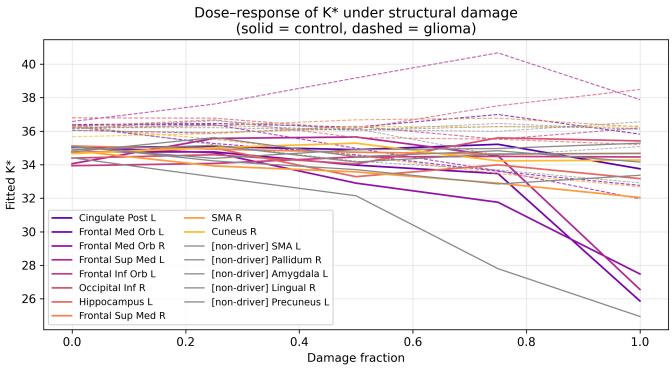

FIG. 10. Dose-response of fitted coupling under structural damage. Solid lines show K^* values obtained when fitting to the control PLI target, and dashed lines show fits to the glioma target, for the 10 virtual-lesioning analysis driver regions and 5 non-driver comparison regions. Although fitted coupling changes with increasing damage, the trajectories do not show a consistent driver-specific amplification and the control curves do not move toward the glioma-like regime.

on the Hopf + PLI + single-K stack inherited from the original paper. Third, both the reproduction and the extension are based on static, phase-based functional connectivity from PLI, and therefore do not capture temporal fluctuations or more complex metastable dynamics [3]. Fourth, optimal couplings were read from a 60-point K grid at fixed λ; per-region differences smaller than the resulting $\sim$0.8 quantisation should be treated as ordinal rather than absolute, and a joint (K, λ) sweep at every lesion or damage level was not performed. Fifth, the AAL78 parcellation is coarse for some regions (e.g. precuneus is one ROI; cerebellum is excluded), which can affect per-region rankings. Sixth, the structural-perturbation analysis tested 15 of 78 regions, so the driver vs non-driver Mann–Whitney comparison has limited statistical power and a whole-brain extension is the obvious next step. Finally, the virtual-lesioning atlas is a model-based measure of how strongly masking a region changes fitted coupling, and should not be interpreted as a direct causal map of tumour influence in vivo; the same caution applies to the structural-perturbation extension, where scaling all incoming and outgoing edges of a node by $(1 - d)$ is only one idealised damage operator and the negative result does not rule out more selective or biologically grounded perturbations producing different outcomes.

4.6. Future directions

Future work should test more biologically specific perturbation schemes rather than uniformly scaling all incoming and outgoing connections of a region. For example, perturbations could target tumour-adjacent edges, long-range fibres, or connections weighted by empirical lesion maps. The model could also be extended beyond

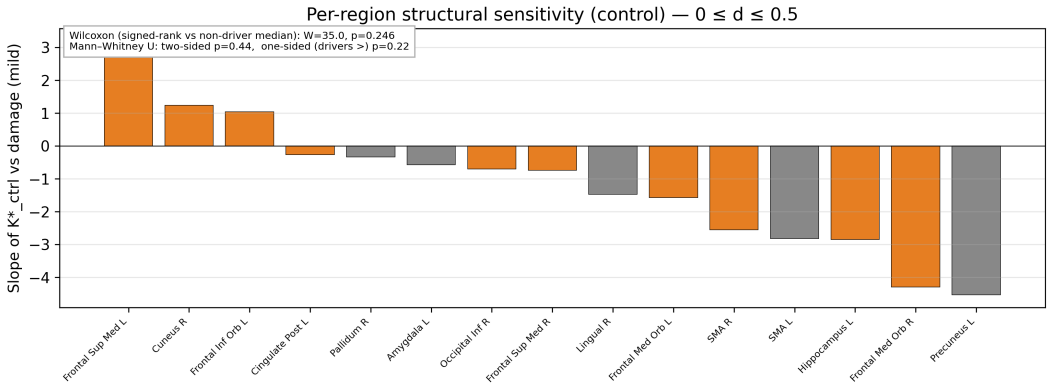

FIG. 11. Per-region structural sensitivity of the control fit over the mild damage range ($0 \leq d \leq 0.5$). Bars show the slope of K^*_{control} versus damage for each perturbed region, ranked from largest to smallest, with driver regions highlighted. Contrary to the original hypothesis, driver regions do not separate clearly from the non-driver set by exhibiting systematically steeper damage responses.

global parameters by allowing region-specific changes in coupling or excitability. Finally, applying the framework to subject-level PLI and structural connectivity would allow testing whether region-wise lesioning effects predict individual tumour location, symptom profile, or clinical outcome.

CONCLUSION

We qualitatively reproduced the main findings of Alexandersen et al.: the Hopf whole-brain model fits were organised by the normalised coupling $\kappa = K/\sqrt{\lambda}$, the glioma target required higher effective coupling than the control target, and tumour-region functional connectotomy reduced fitted coupling. These results support the original interpretation that glioma changes the modelled balance between local oscillatory dynamics and interregional coupling.

Our extensions asked whether this connectotomy effect could be localised to specific regions and whether those regions should be interpreted as anatomical damage drivers. Region-wise virtual lesioning showed that fitted-coupling sensitivity is spatially heterogeneous and partly associated with eigenvector centrality. However, direct structural perturbation gave a different result: weakening the connections of top virtual-lesioning drivers did not move control fits toward the glioma-like higher-coupling regime, and virtual-lesioning strength did not predict structural-perturbation sensitivity.

The main conclusion is therefore that functional connectotomy identifies regions that influence the simulated–empirical PLI fitting objective, not necessarily regions whose anatomical disconnection causes glioma-like dynamics. In this model, functional fit-sensitivity and anatomical damage sensitivity are separable. Thus, while glioma is associated with a shift toward stronger effective interregional coupling, that shift cannot be explained by a simple structural-disconnection manipulation of the identified functional drivers.

5. AUTHOR CONTRIBUTIONS

G.M., L.O., B.R., and L.Q.L. contributed equally to conceptualisation, analysis, and manuscript preparation.

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* All figures were generated using custom Python code written by the authors. The full analysis and figure-generation is available at `https://github.com/BOBRICE1/Functional-connectotomy`.

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